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CASE REPORT

Bilateral central corneal keratolysis (corneal melt) in sarcoidosis

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ABSTRACT

Background and aim: Corneal involvement in sarcoidosis is rare, and central corneal keratolysis has not been previously described. We present the first reported case of bilateral central corneal keratolysis in a patient with multi-organ sarcoidosis.
Methods: Case report
Results: A 37-year-old woman with pulmonary, cutaneous, and ocular sarcoidosis presented with severe bilateral ocular pain, photophobia, and rapidly progressive vision loss. Examination revealed severe uveitis with bilateral hypopyons and central corneal keratolysis. Results from infectious, rheumatological, and vasculitis work-up were negative. She was treated with high-dose intravenous corticosteroids, methotrexate, and infliximab but developed significant stromal scarring with persistent visual impairment.
Conclusions: This case highlights the need to consider sarcoidosis in the differential diagnosis of rapidly progressive keratolysis, particularly in patients with known systemic disease or anterior sarcoid uveitis.

Key words: ocular sarcoidosis, central corneal keratolysis, corneal melt, anterior uveitis, facial piercings, tattoo sarcoidosis, hypopyon

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Introduction

Ocular sarcoidosis most commonly involves the uveal tract (1). Corneal involvement is rare and only a few cases of interstitial keratitis, band keratopathy (1, 2), and peripheral ulcerative keratitis (PUK) have been described (3, 4). We present what is, to our knowledge, the first reported case of bilateral central corneal keratolysis (corneal melt) in a patient with systemic sarcoidosis.

Case description

Our patient is a 37-year-old Black female with pulmonary, cutaneous, and ocular sarcoidosis (left anterior uveitis) who presented with one week of severe bilateral ocular pain, redness, photophobia, and vision loss. There was no preceding ocular trauma or eye infection. Her sarcoidosis was diagnosed nine years earlier via bronchoscopy and her systemic disease was previously well controlled on prednisone 5 mg daily and hydroxychloroquine 200 mg twice daily. She had however, self-discontinued her medications the year prior for financial reasons.

Eye examination revealed bilateral circumcorneal injection, along with keratic precipitates and bilateral hypopyons (Figures 1A and 1B). She had a longstanding left eyebrow (Figure 1A) and right nasal piercing (Figure 1B), as well as multiple cutaneous tattoo-associated papules which had reoccurred about four weeks prior to her acute presentation (Pictures not shown). Visual acuity was counting fingers in the right eye (OD) and hand motion in the left eye (OS). Slit lamp examination revealed diffuse injection OS along with bilateral ciliary flush and hypopyons (Figures 2A and 2B) which were quantified as 1.5 mm in the right eye (Figure 2A) and 3.5 mm in the left eye (Figure 2B). The corneas exhibited large central epithelial defects (6 x 9.5 mm OD as shown in figure 2C and 7 x 9.5 mm OS as shown in figure 2D) with hazy borders, stromal edema, and corneal thinning OS>OD. No corneal infiltrates were observed in either eye. Bilateral pupillary reactions were normal, and the patient was phakic with clear lens bilaterally. The view of the fundus was obscured, however, B-scan ultrasonography revealed clear bilateral vitreous chambers.

She was diagnosed with severe bilateral anterior uveitis complicated by bilateral central corneal keratolysis and admitted for urgent immunosuppression. Extensive infectious and rheumatological work-up including antineutrophil cytoplasmic antibody (ANCAs) titers was negative (Table 1).

She was initiated on pulse dose intravenous methylprednisolone (1 g daily for 5 days) with subsequent addition of methotrexate and infliximab (5 mg/kg).

Figure 1. Showing severe circumcorneal injection (fat arrow/Figure 1A) and keratic precipitates (thin arrow /Figure 1B) in the left eye (OS). Left eyebrow (Figure 1A) and right nasal piercing (Figure 1B) are noted. She noted that both piercings had been performed at least 5-years previously and neither rings had been changed recently.

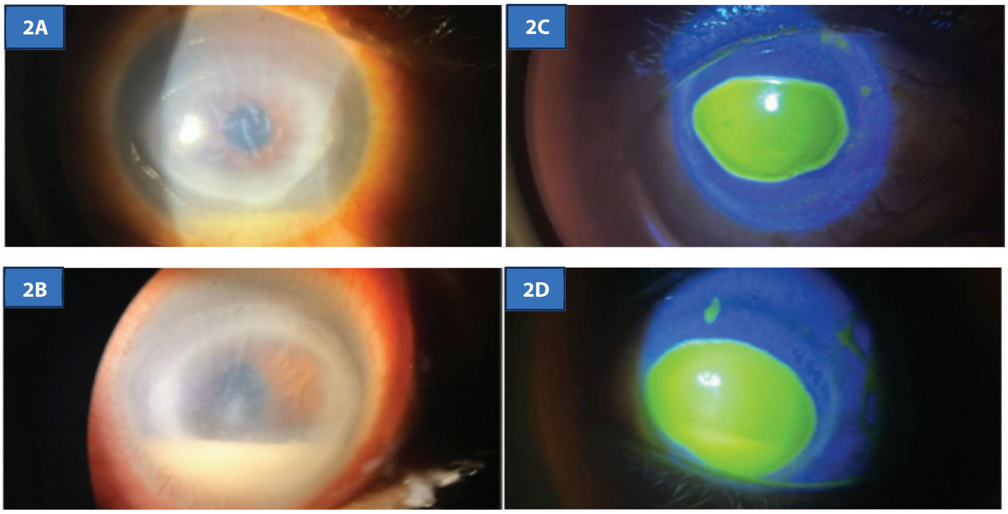

Figure 2. Slit lamp examination showed bilateral hypopyons and central corneal melts with hazy borders and stromal edema. (A) The right eye and (B) left eye. Fluorescein staining showed (C) large central epithelial defects in the right eye and (D) left eye. The left eye was more severely affected than the right eye.

Table 1. Summary of work-up.

Test	Result/ Comments
Autoimmune/Vasculitis Work-up	
ANA with reflex to ENA, DSDNA, RNP, SM, SS-A, SS-B	Negative
Anti-DNA (DS) Antibodies	Negative
Sjogren's Anti-SS-A & Anti-SS-B antibodies	Negative
Rheumatoid Factor	Negative
MPO and PR3 ANCAs	Negative
Infectious Work-up	
QuantiFERON-TB Gold Plus	Negative
Lyme disease serology	Negative
Syphilis Antibodies	Negative
HIV Ag/Ab Combined assay	Negative
Blood cultures (bacterial and fungal)	Negative
Acute Hepatitis A, B, C serology	Negative
Others:	Ref. Range
CRP 4.2 mg/dL	< 5mg/dL
Vit D 1-25-diOH - 29.2 pg/mL	24.8-81.5 pg/mL

Abbreviations: ANA = Antinuclear Antibody, ENA = Extractable Nuclear Antigen, DSDNA = double stranded DNA, RNP = Ribonucleoprotein, SM = Smith Antibody, Anti-SSA (anti-Ro) & Anti-SSB (Anti-La) antibodies, HIV = Human Immunodeficiency Virus, Ag/Ab = antigen and antibody combined immunoassay, MPO = Myeloperoxidase, ANCA = Antineutrophil cytoplasmic antibody, PR3 = Proteinase 3, CRP = C-reactive protein, Vit D 1-25-diOH = 1, 25-dihydroxyvitamin D (Calcitriol).

Empiric therapy for bacterial and viral keratitis was also initiated. Doxycycline and vitamin C were added to reduce metalloprotease-mediated corneal stromal degradation, and both corneas were stabilized using cyanoacrylate glue and bandage contact lenses.

She was discharged one week later, on oral prednisone taper, methotrexate, and infliximab (5 mg/kg every 4 weeks). At her one-month follow-up, she had developed bilateral neurotrophic corneas however, the stromal haze had improved (OD > OS), and the corneal epithelium OD had healed. She had residual defect OS. Scleral lenses subsequently improved vision OD (to 20/80) but were ineffective in the left eye (maximal improvement was to 20/400) due to significant stromal scarring. She subsequently underwent deep anterior lamellar keratoplasty OS one year later but continues to have a posterior residual stromal haze which has prevented visual improvement beyond 20/400. A penetrating keratoplasty is planned for the OS.

Discussion

Sarcoidosis ocular manifestations most commonly involve the uveal tract (1). The cornea is rarely affected perhaps because of its relative avascularity (1, 5). A few cases of peripheral ulcerative keratolysis (PUK) have been reported (3, 4) however, we believe this to be the first reported case of central corneal keratolysis.

The peripheral cornea has a higher concentration of Langerhans cells, immunoglobulins and complement components and is more susceptible to keratolysis which occurs as a result of autoimmune vasculitis of the limbal vessels (6). Conversely, central cornea keratolysis is not mediated by vasculitis and is thought to occur from excessive matrix metalloproteinase (MMP) activity triggered by diverse stimuli (such as neurotrophic epithelial failure, severe dry eyes, and topical non-steroidal anti-inflammatory drug toxicity) (7, 8).

We believe the patient's response to high-dose immunosuppression, as well as the absence of other identifiable infectious, autoimmune, or inflammatory causes following an extensive work-up supports an immune-mediated sarcoidosis-related mechanism. We speculate that her central corneal keratolysis was probably triggered by severe anterior uveitis which is known to disrupt epithelial barrier integrity and limbal stem cell function, exposing the stroma to tear film-derived proteases and inflammatory mediators. Furthermore, we suspect that the presence of large hypopyons likely introduced activated neutrophils with release of elastase, reactive oxygen species, and additional MMPs further amplifying stromal degradation and keratolysis (9).

Our patient was aggressively treated with pulse dose intravenous steroids, methotrexate, and infliximab. There are no guidelines for the treatment of corneal keratolysis (corneal melt) in sarcoidosis, and previously reported cases of PUK and interstitial keratitis were treated with high-dose steroids, oral Cytoxan and topical cyclosporine (4). Data from patients with rheumatoid arthritis suggest that early initiation of immunotherapy with anti-metabolites and anti-tumor necrosis factor (anti-TNF) is associated with improved visual outcomes in patients with corneal involvement(10). Despite these interventions in our patient however, significant stromal scarring ultimately occurred necessitating deep anterior lamellar keratoplasty OS with persistent severe visual impairment. This underscores the vision-threatening potential of this rare presentation and emphasizes the need to raise awareness of its possible occurrence.

Conclusion

In conclusion, we present the case of a 37-year-old Back Female with uncontrolled systemic sarcoidosis, who presented with severe sight-threatening uveitis and bilateral central corneal keratolysis (corneal melt) with visual loss despite aggressive immunosuppressive therapy. To our knowledge, this is the first reported case of bilateral central corneal keratolysis in sarcoidosis.

Authors' Contribution: All the authors were involved in the design, writing and editing of this manuscript.

Declaration on the Use of AI: None.

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangement etc.) that might pose a conflict of interest in connection with the submitted article.

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Consent for Publication: Written informed consent was obtained from the patient for publication of this case report and any accompanying images.

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